

MGT404: Financial Econometrics and Empirical Methods (946331)

Master in Quantitative Finance 2025-26, UCSD
8:00AM-10:50 AM PST

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OFFICE HOURS: TBA or by appointment

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COURSE DESCRIPTION

This course is designed to help students bridge the gap between theoretical financial models and the world of empirics and financial econometrics. The 10-week class will take students through the major accomplishments (and failures) of empirical finance and financial econometrics during the last quarter of a century. Students will have to read some of the truly seminal works in financial econometrics. Empirical exercises will require students to write their own procedures and to use real market data. These exercises will help us all truly internalize the content of the course. Upon completion of this course, students should:

- Have a good picture of the main financial econometrics techniques
- Have a good understanding of the important topics in empirical finance
- Be able to implement many of the widely used methods in modern empirical finance
- Have an in-depth knowledge in an area of their choice.

Prerequisites:

- A quantitatively based “Investments” class is a prerequisite for the course.
- Familiarity with computers. Students should be able to download and manipulate data.
- Knowledge of a programming language or statistical package (e.g., Python, Matlab, R) is required. Knowledge of “canned” statistical programs such as Eviews, SAS, or Stata is not enough. Students will be expected to build their own library of routines and procedures from the ground up.

SCHEDULE

Classes are held once a week on **Tuesdays from 8:00AM-10:50 AM PST** in **WFH 4N128**.
Important: Class attendance is **mandatory**.

Office hours will be held in-person or via Zoom. If you cannot make it during office hours, please e-mail me about an appointment. I welcome questions via email.

CANVAS

Please familiarize yourself with Canvas. Each week, I will post all materials for the class – lecture slides, assignments, quizzes, additional material, Python code and others — in a Module.

COURSE MATERIALS

The weekly lecture notes will be available on Canvas, posted in the morning of the lecture.

There is one required book as well as a few suggested texts:

Required Book:

1. J.Y. Campbell, A.W. Lo and A.C. MacKinlay, *The Econometrics of Financial Markets*, Princeton University Press, Princeton, 1997.

Very Helpful Text:

1. J.D. Hamilton, *Time Series Analysis*, Princeton University Press, 1994.

ASSIGNMENTS AND GRADING

The grading policy is as follows:

1. Written/Computer Exercises (40%) – should be done in pre-assigned groups of no more than 5 people. One written assignment per group is to be turned in on time.
2. Weekly Quizzes (10%) – We will assign weekly quizzes after each lecture on Canvas. They will help you keep up with the material.
3. Participation and/or peer evaluations (10%)
4. Final Examination (40%)—The final examination may involve manipulation of data. It will be administered during Finals week. It may be in-person or online.

To prevent one bad performance from affecting your grade, I will drop the lowest written exercise and weekly quiz grade.

GETTING TO KNOW YOU

High-paced lectures are not the ideal platform for interaction. I like to get to know my students.

I will schedule 20 mins appointments with each group outside of normal lecture hours. During these appointments, I would like to get to know you better. Please be prepared to say a few words about yourself -- your background, your interests, and your goals – anything that will help me get to know you. These “get to know you” appointments will be scheduled during weeks 2-7. Groups will be able to pick a time on a sign-up sheet. The appointments can be done either in-person or via Zoom. First-come, first-served!

GENERATIVE AI POLICY

You can use ChatGPT or other generative AI tools to complete the coding or other parts of the assignments. However, you are responsible for the final version of your assignments. Excuses of the sort “I got it wrong because of ChatGPT” or “I copied the answer from ChatGPT even though I cannot understand it” will not be accepted.

DETAILED SYLLABUS

We may deviate from the schedule depending on the pace the class feels comfortable with.

Week 1	Sep 30:	Introduction to class; Basics of probability, statistics and econometrics. Introduction to time series
Week 2	Oct 07:	Time series; Ordinary Least Squares (OLS). Asymptotic and small sample properties
Week 3	Oct 14:	Estimation of AR models and predictive regressions (basics); Predictive regressions (in depth); Tests of Market Efficiency
Week 4	Oct 21:	May be on Zoom on a date and time other than the usual. To be announced in class--Cross sectional regressions (basics); Cross sectional regressions (in depth); Endogeneity
Week 5	Oct 28:	The CAPM and Multifactor models; Tests of the CAPM and Multifactor models; Fama-French and Fama-MacBeth regressions
Week 6	Nov 04:	Vector Autoregressions (VARs); Maximum Likelihood Estimation (MLE); Introduction to volatility modeling and estimation
Week 7	Nov 11:	Veteran's Day. May be on Zoom on a date and time other than the usual. To be announced in class. Realized Volatility models and estimation; ARCH/GARCH models and estimation
Week 8	Nov 18:	Other volatility models (implied volatility, MIDAS, ...); The importance of anchoring returns to economic fundamentals--simple Gordon Growth and Consumption CAPM (CCAPM); Spurious Regressions and non-stationary time-series (DF test, cointegration)
Week 9	Nov 25:	Empirical market microstructure; Bootstrapping and simulating distributions; Introduction to Generalized Method of Moments (GMM)
Week 10	Dec 02:	Advanced (if time allows) – Intro to Machine learning, Kalman Filtering; Review
FINAL	Dec 09:	Comprehensive. In-person (8:00am-11:00am in OTRSN 1E106) or Online.

Other topics that we will cover as time permits: Continuous Time Processes. Estimation. Testing. Non-Parametric Estimation. Estimation of Quantiles. Kernels and Density Estimators. Non-Parametric Estimation of State-Price Densities and Non-Parametric Pricing of Interest Rate Derivatives.

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind. The complete UCSD Policy on Integrity of Scholarship can be viewed at:

<http://wwwsenate.ucsd.edu/manual/appendices/app2.htm#AP14>

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member. The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.** Please visit the [OSD website](#) for further information or contact the Office for Students with Disabilities at (858) 534-4382 or osd@ucsd.edu.